ADCS Homework 4

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I. Actuator Specifications

The ISS uses two classes of ADCS actuators: four large double-gimbal control moment gyros (CMGs) on the Z1 truss for primary non-propulsive attitude control, and the Zvezda service-module RCS cluster for momentum desaturation and reboost. Public information on the exact mounting geometry of the CMGs is incomplete, so we model the array as the standard symmetric 4-CMG pyramid used throughout the literature [1], [2], and approximate each CMG as a single-gimbal unit at the cluster level (which is how the Bedrossian et al. analyses treat them for momentum-envelope and steering-law design).

Both actuator groups are instantiated through the `mujoco_orbit` public API as `ControlMomentGyroSpec` and `ThrusterSpec`, compiled into `MjoModel` against `iss_model.xml`. All specifications, geometry, and Jacobians below are produced by `ISS/hw4/actuator_specs.py`.

The body frame convention is LVLH-aligned at rest: $+X$ along-track (roll), $+Y$ cross-track (pitch), $+Z$ nadir (yaw). The right-handed LVLH attitude used in Sec. II places $+Y_b$ opposite the simulator's positive orbit-normal axis.

A. CMG Cluster

a) Device-level specifications.: Each CMG is a Honeywell M1000-class unit with the parameters summarized in Table I [1], [3]. The published rotor-momentum magnitude is $h = 4760\,\mathrm{N\,m\,s}$ and the peak output torque is $\tau_{\max} = 258\,\mathrm{N\,m}$; the maximum gimbal rate follows from these as $\dot{\theta}_{\max} = \tau_{\max}/h \approx 0.054\,\mathrm{rad/s} \approx 3.1\,\mathrm{deg/s}$.

TABLE I
PER-CMG SPECIFICATIONS (HONEYWELL M1000).

Quantity	Value
Rotor angular momentum h	$4760\,\mathrm{N\,m\,s}$
Peak output torque $\tau_{\max}$	$258\,\mathrm{N\,m}$
Max gimbal rate $\dot{\theta}_{\max}$	$0.054\,\mathrm{rad/s}$ ($3.11\,\mathrm{deg/s}$)
Gimbal angle range	full rotation (wrap-around)

b) Pyramid array geometry.: Four CMGs are mounted on the Z1 truss with gimbal axes arranged in the standard symmetric pyramid with skew angle $\beta = \cos^{-1}(1/\sqrt{3}) = 54.7356°$, which maximizes the inscribed momentum sphere [1]. We orient the pyramid apex along $+Z_{\text{body}}$. With azimuths $\phi_i \in \{0,\ \pi/2,\ \pi,\ 3\pi/2\}$, the gimbal axes $\hat{\boldsymbol{g}}_i$, the spin axes at neutral gimbal $\hat{\boldsymbol{s}}_i(0)$, and the output (torque) axes $\hat{\boldsymbol{t}}_i(0) = \hat{\boldsymbol{g}}_i \times \hat{\boldsymbol{s}}_i(0)$ are listed in Table II. The neutral spin axes are chosen so that the net array momentum vanishes at $\boldsymbol{\theta} = \mathbf{0}$:

$$\boldsymbol{H}(\mathbf{0}) = \sum_{i=1}^{4} h\,\hat{\boldsymbol{s}}_i(0) = \mathbf{0}. \tag{1}$$

TABLE II
CMG GEOMETRY IN THE BODY FRAME ($\beta = 54.7356°$, APEX $\parallel +Z_{\text{BODY}}$).

CMG	$\hat{\boldsymbol{g}}_i$	$\hat{\boldsymbol{s}}_i(0)$	$\hat{\boldsymbol{t}}_i(0)$
1 ($\phi = 0$)	$(+0.816,\ 0,\ +0.577)$	$(0,\ +1,\ 0)$	$(-0.577,\ 0,\ +0.816)$
2 ($\phi = \pi/2$)	$(0,\ +0.816,\ +0.577)$	$(-1,\ 0,\ 0)$	$(0,\ -0.577,\ +0.816)$
3 ($\phi = \pi$)	$(-0.816,\ 0,\ +0.577)$	$(0,\ -1,\ 0)$	$(+0.577,\ 0,\ +0.816)$
4 ($\phi = 3\pi/2$)	$(0,\ -0.816,\ +0.577)$	$(+1,\ 0,\ 0)$	$(0,\ +0.577,\ +0.816)$

c) Cluster momentum and Jacobian.: For an SGCMG at gimbal angle θ_i, the spin axis in the body frame is

$$\hat{\boldsymbol{s}}_i(\theta_i) = \cos\theta_i\,\hat{\boldsymbol{s}}_i(0) + \sin\theta_i\,\hat{\boldsymbol{t}}_i(0). \tag{2}$$

The cluster angular-momentum map and the body-frame torque Jacobian with respect to commanded gimbal rates $\dot{\boldsymbol{\theta}} \in \mathbb{R}^4$ are therefore

$$\boldsymbol{H}(\boldsymbol{\theta}) = \sum_{i=1}^{4} h\,\hat{\boldsymbol{s}}_i(\theta_i), \tag{3}$$

$$A(\boldsymbol{\theta}) = \frac{\partial \boldsymbol{\tau}_{\text{body}}}{\partial \dot{\boldsymbol{\theta}}} = -h\left[\hat{\boldsymbol{t}}_1(\theta_1) \mid \hat{\boldsymbol{t}}_2(\theta_2) \mid \hat{\boldsymbol{t}}_3(\theta_3) \mid \hat{\boldsymbol{t}}_4(\theta_4)\right], \tag{4}$$

where $\hat{\boldsymbol{t}}_i(\theta_i) = \hat{\boldsymbol{g}}_i \times \hat{\boldsymbol{s}}_i(\theta_i)$. The negative sign in (4) comes from Newton's third law: the body torque is the reaction to $\mathrm{d}\boldsymbol{h}_i^{\text{body}}/\mathrm{d}t = h\,\dot{\theta}_i\,\hat{\boldsymbol{t}}_i$.

Evaluated at the neutral gimbal state $\boldsymbol{\theta} = \mathbf{0}$,

$$A(\mathbf{0}) = \begin{bmatrix} +2748 & 0 & -2748 & 0 \\ 0 & +2748 & 0 & -2748 \\ -3887 & -3887 & -3887 & -3887 \end{bmatrix} \mathrm{N\,m/rad/s}. \tag{5}$$

Rows 1 and 2 have the sparsity pattern $\pm h\cos\beta \cdot \hat{\boldsymbol{s}}_i^{xy}$ (only opposing-azimuth CMGs contribute to roll/pitch torque at $\boldsymbol{\theta} = \mathbf{0}$), and the row-3 entries are all $-h\sin\beta = -3887$ N·m per rad/s, so all four CMGs contribute constructively to yaw torque in the zero-gimbal configuration. This is a well-known feature of the pyramid array and is why the "null motion" direction at $\boldsymbol{\theta} = \mathbf{0}$ ($\dot{\boldsymbol{\theta}} = (1,1,1,1)$) produces a pure $-Z$ torque of magnitude $4h\sin\beta \cdot \dot{\theta}_{\max} \approx 842\,\mathrm{N\,m}$.

d) Momentum envelope.: Sampling $\boldsymbol{\theta} \in [-\pi, \pi]^4$ uniformly and plotting $\boldsymbol{H}(\boldsymbol{\theta})$ gives the solid region shown in Fig. 1. Its maximum extent is $4h = 19\,040\,\mathrm{N\,m\,s}$ (when all spin axes align), and the largest inscribed sphere has radius $2h\cos(\beta/2) \approx 8454\,\mathrm{N\,m\,s}$.

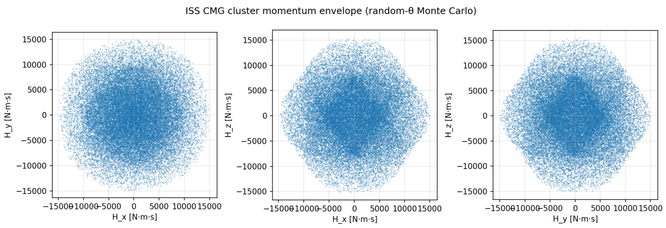

Fig. 1. Cluster angular-momentum envelope of the 4-CMG pyramid (40 000 random gimbal configurations). Projections on the three body-frame coordinate planes. The envelope is convex in each projection; internal singular surfaces exist within the solid region (not shown).

e) Output-torque set.: At $\boldsymbol{\theta} = \mathbf{0}$, the image of the unit $\dot{\boldsymbol{\theta}}$-ball under $A(\mathbf{0})$, scaled by $\dot{\theta}_{\max}$, is the rank-3 ellipsoid of body torques simultaneously achievable without saturating any gimbal (Fig. 2). The ellipsoid is stretched along $\pm Z$ because row 3 of (5) is coherent across all four CMGs.

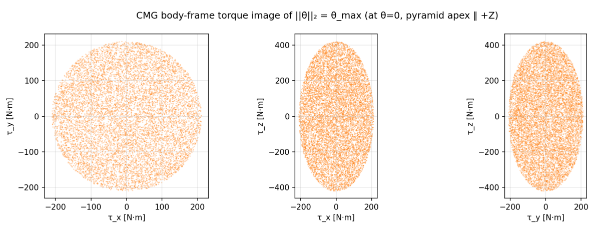

Fig. 2. Image of $\|\dot{\boldsymbol{\theta}}\|_2 = \dot{\theta}_{\max}$ under $A(\mathbf{0})$ — the body-frame torques reachable without exceeding any gimbal-rate limit at the zero-gimbal configuration.

B. Zvezda RCS Thrusters

The Russian Zvezda service module provides the ISS's only on-station propulsive ADCS authority. Its aft propulsion assembly (PAO) houses ~32 small DPO attitude thrusters (130 N class) [4] and 2 large DKD reboost engines (3070 N class). For this analysis we use a simplified symmetric layout of 8 attitude thrusters on the aft rings plus the 2 aft-facing reboost engines (Table III). Positions are given with respect to the ISS center of mass, which is approximately 30 m forward of Zvezda's aft bulkhead.

a) Wrench Jacobian.: Each thruster produces a body-frame force $\hat{\boldsymbol{d}}_i F_i$ and a torque $\boldsymbol{r}_i \times \hat{\boldsymbol{d}}_i F_i$ about the CM, so the full $6 \times N$ wrench Jacobian is

$$B = \begin{bmatrix} \hat{\boldsymbol{d}}_1 & \cdots & \hat{\boldsymbol{d}}_N \\ \boldsymbol{r}_1 \times \hat{\boldsymbol{d}}_1 & \cdots & \boldsymbol{r}_N \times \hat{\boldsymbol{d}}_N \end{bmatrix}, \qquad \begin{bmatrix} \boldsymbol{F} \\ \boldsymbol{\tau} \end{bmatrix}_{\text{body}} = B\,\boldsymbol{u}, \quad (6)$$

with $\boldsymbol{u} \in [0, F_{\max,i}]^N$. The numerical evaluation gives

$$B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & +1 & +1 \\ +1 & -1 & +1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & +1 & +1 & -1 & -1 & 0 & 0 \\ -3 & +3 & +3 & -3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & +29 & +31 & -29 & -31 & 0 & 0 \\ -30 & +30 & -30 & +30 & 0 & 0 & 0 & 0 & -0.8 & +0.8 \end{bmatrix}. \quad (7)$$

The top three rows give body-frame force per unit thrust and the bottom three give torque per unit thrust. Firing each pair of Y-thrusters in the same direction produces a pure roll or yaw couple; firing each Z-pair in the same direction produces a pure pitch couple; the two DKD engines provide pure along-track force (with a small residual yaw couple from their $\pm Y$ lever arms).

b) Maximum body wrenches.: Saturating the thrusters that contribute in a given axis direction gives the upper-bound pure-torque capabilities in Table IV. Roll authority is the weakest (limited to the ± 3 m ring radius), while yaw authority is the strongest (benefiting from the ~ 30 m lever arm along $-X$).

c) Use in the overall ADCS.: Although the thrusters have much higher instantaneous torque than the CMG cluster, they are propellant-limited and are not used for continuous attitude regulation. Their role is to perform CMG desaturation (momentum dumping) when secular disturbance torques accumulate to the cluster momentum limit, and to execute orbit reboost. Quantitative estimates of how often desaturation is required are addressed in Sec. II.

II. Environmental Torques

For HW4 Q2 I added gravity-gradient torque and aerodynamic drag to the single-rigid-body ISS model. Drag is the appropriate surface-force perturbation for this orbit: at 408 km, the atmosphere model gives $\rho = 2.28 \times 10^{-13}\,\mathrm{kg\,m^{-3}}$ and $v = 7.66\,\mathrm{km/s}$, so $q = \frac{1}{2}\rho v^2 = 6.71 \times 10^{-6}\,\mathrm{N\,m^{-2}}$. With $C_D = 2.2$ the effective drag pressure is $1.48 \times 10^{-5}\,\mathrm{N\,m^{-2}}$, larger than a representative solar-radiation pressure load ($1.8P_{\odot} = 8.21 \times 10^{-6}\,\mathrm{N\,m^{-2}}$). Drag also acts continuously in LEO and changes the orbit energy, so it is the more relevant choice for the ISS than SRP.

All Q2 data are generated by `ISS/hw4/environmental_torques.py`. The script uses

$$J = \mathrm{diag}(128,\ 107,\ 201) \times 10^6\ \mathrm{kg\,m^2},$$

the HW2 flat-plate ISS surface model, and a circular 408 km, 51.6° orbit with period 92.72 min. The simulator LVLH axes are $+x$ radial-out, $+y$ along-track, and $+z$ orbit-normal. The right-handed nominal ISS attitude used here is computed as

$$+X_b \parallel +y_{\text{LVLH}}, \qquad +Z_b \parallel -x_{\text{LVLH}}, \qquad +Y_b \parallel -z_{\text{LVLH}}.$$

This is the "nominal alignment" used in the plots and tables.

A. Torque Models

For each rigid body, the gravity-gradient torque is

$$\boldsymbol{\tau}_{\text{gg}} = \frac{3\mu}{r^3}\,\hat{\boldsymbol{r}} \times (J\hat{\boldsymbol{r}}), \quad (8)$$

where $\hat{\boldsymbol{r}}$ and J are expressed in the same frame. The code rotates MuJoCo's principal-axis inertia into the world/LVLH frame using `ximat`, so non-diagonal `fullinertia` models are handled correctly. This is the missing term for a large spacecraft modeled as one rigid body; point gravity at the body COM cannot generate this rotational torque.

TABLE III
SIMPLIFIED ZVEZDA RCS LAYOUT MODELED IN ACTUATOR_SPECS.PY.

Role	Label	Position [m]	Direction	$F_{\max}$ [N]
Attitude (ring, top/bot ±Y)	ACS_+Y_top	$(-30, +3, +3)$	$(0, +1, 0)$	130
	ACS_-Y_top	$(-30, -3, +3)$	$(0, -1, 0)$	130
	ACS_+Y_bot	$(-30, +3, -3)$	$(0, +1, 0)$	130
	ACS_-Y_bot	$(-30, -3, -3)$	$(0, -1, 0)$	130
Attitude (ring, fwd/aft ±Z)	ACS_+Z_fwd	$(-29, 0, +3)$	$(0, 0, +1)$	130
	ACS_+Z_aft	$(-31, 0, +3)$	$(0, 0, +1)$	130
	ACS_-Z_fwd	$(-29, 0, -3)$	$(0, 0, -1)$	130
	ACS_-Z_aft	$(-31, 0, -3)$	$(0, 0, -1)$	130
Reboost (DKD)	DKD_stbd	$(-31.5, +0.8, 0)$	$(+1, 0, 0)$	3070
	DKD_port	$(-31.5, -0.8, 0)$	$(+1, 0, 0)$	3070

TABLE IV
UPPER-BOUND PURE-TORQUE AND REBOOST CAPABILITIES FROM THE SIMPLIFIED ZVEZDA LAYOUT.

Command	Max magnitude
Pure roll torque τ_x	780 N m
Pure pitch torque τ_y	7800 N m
Pure yaw torque τ_z	10 256 N m
Pure $+X$ force (reboost)	6140 N

Drag is applied to each flat plate as

$$\boldsymbol{F}_D = -\frac{1}{2}\rho C_D A_{\text{proj}} \|\boldsymbol{v}_{\text{rel}}\| \, \boldsymbol{v}_{\text{rel}}, \qquad \boldsymbol{\tau}_D = \boldsymbol{r}_{\text{cp}} \times \boldsymbol{F}_D, \tag{9}$$

where $\boldsymbol{v}_{\text{rel}}$ includes atmospheric co-rotation.

B. Experiment Design

The homework asks for free simulations over several orbits, so the main experiment is a three-orbit uncontrolled response with four perturbation sets: none, gravity gradient only, drag only, and both. I run that experiment from two initial attitudes:

- exact nominal LVLH/nadir alignment; and
- an off-nominal body-axis offset of $(5°, -3°, 2°)$, whose total attitude error is $6.12\,\text{deg}$.

The second initial condition is not a hidden controller assumption; it is a stated perturbation that makes the gravity-gradient dynamics visible. Separately, I compute fixed-reference torque histories over one day. Those histories estimate how much angular momentum the CMGs would accumulate if an attitude controller were holding a commanded attitude.

C. Torque Magnitudes

The nominal LVLH attitude has zero gravity-gradient torque because the radial direction is exactly the principal $-Z_b$ axis: $\hat{\boldsymbol{r}} \times J\hat{\boldsymbol{r}} = 0$. This is why the gravity-gradient bar is effectively absent in the nominal case. When the spacecraft is started from the stated off-nominal attitude, gravity gradient is the dominant torque. The analytical all-attitude upper bound is

$$\tau_{\text{gg,max}} = \frac{3\mu}{2r^3}\left(I_{\max} - I_{\min}\right) = 179.84\,\text{N m}.$$

TABLE V
FIXED-REFERENCE ENVIRONMENTAL TORQUES USED FOR SIZING.

Reference attitude	Max $\|\boldsymbol{\tau}\|$	Orbit-avg. $\|\bar{\boldsymbol{\tau}}\|$
Nominal GG	≈ 0	≈ 0
Nominal drag	6.11×10^{-3} N m	2.79×10^{-4} N m
Offset GG	34.32 N m	34.25 N m
Offset drag	9.12×10^{-3} N m	5.83×10^{-3} N m

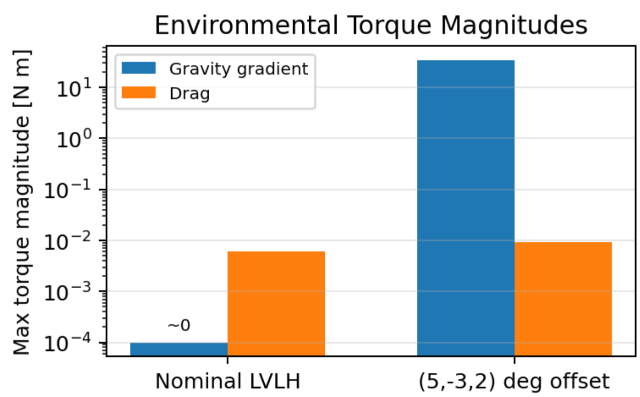

Fig. 3. Maximum fixed-reference torque magnitudes. The nominal gravity-gradient value is numerical roundoff because the radial direction is a principal inertia axis.

D. Orbit Averages and Momentum Accumulation

For the nominal LVLH reference attitude, gravity gradient contributes no secular momentum. The drag orbit-average is

$$\bar{\boldsymbol{\tau}}_D = \left(-8.23 \times 10^{-8},\ -5.16 \times 10^{-6},\ 2.79 \times 10^{-4}\right)\,\text{N m},$$

which gives $28.5\,\text{N m s}$ maximum accumulated momentum over one day in the time-domain integration. The orbit-average estimate gives $24.1\,\text{N m s/day}$. Both are tiny compared with the $8454\,\text{N m s}$ conservative CMG inscribed momentum radius.

The off-nominal fixed-reference case is different: holding the $(5°, -3°, 2°)$ attitude offset against gravity gradient would accumulate $2.96 \times 10^6\,\text{N m s}$ in one day and would fill the conservative CMG envelope in about $4.1\,\text{min}$. This is a sizing warning, not the expected operating mode.

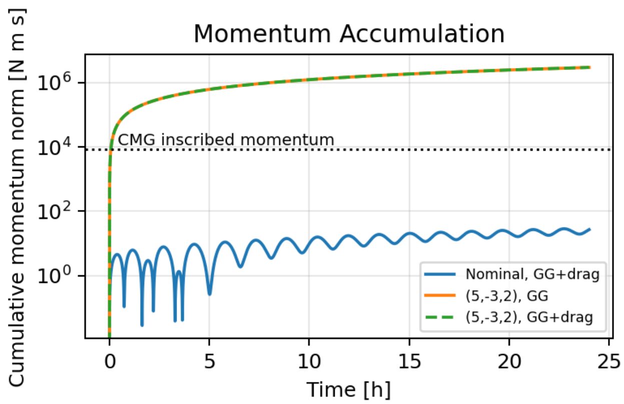

Fig. 4. Cumulative momentum required to reject the modeled disturbances. The nominal case is drag-limited; the offset reference is gravity-gradient dominated, so the gravity-gradient-only and combined curves nearly overlap.

E. Several-Orbit Free Response

Figure 5 shows the free attitude simulations. From exact nominal LVLH, the no-perturbation and gravity-gradient-only cases remain fixed because the initial attitude is an equilibrium. Drag alone causes only $0.011\,\mathrm{deg}$ drift over three orbits. With drag and gravity gradient together, drag first pushes the body away from the exact equilibrium and gravity gradient then dominates the rotational motion, reaching about $180\,\mathrm{deg}$ error and $9.76 \times 10^{-2}\,\mathrm{deg/s}$.

From the off-nominal initial attitude, the no-perturbation case stays at the initial $6.12\,\mathrm{deg}$ error. Drag alone changes that by only $0.38\,\mathrm{deg}$. Gravity gradient drives the main free response, with maximum rate $1.01 \times 10^{-1}\,\mathrm{deg/s}$. The added drag barely changes the curve because the off-nominal gravity-gradient torque is about four orders of magnitude larger than the drag torque.

F. ADCS Implications

The CMG cluster has ample instantaneous torque authority for these modeled disturbances: even the off-nominal $34.32\,\mathrm{N\,m}$ gravity-gradient torque is well below the hundreds of $\mathrm{N\,m}$ available from the CMGs. The main issue is momentum management and attitude stability. In nominal LVLH, the modeled drag load would fill the conservative CMG momentum sphere only after roughly $300\,\mathrm{days}$ to $350\,\mathrm{days}$, depending on whether the one-day peak or the orbit-averaged secular value is used. Real ISS desaturation will be driven by additional unmodeled biases and operational constraints, but this model does not predict frequent momentum dumping from drag alone.

Gravity gradient is not a nominal momentum-dumping driver when the ISS is aligned with LVLH because its torque is zero at that equilibrium. It is still important for the nonlinear attitude simulation: once the vehicle is a few degrees away from the principal-axis alignment, the gravity-gradient torque dominates the free response and must be included in the controller test cases for Q3.

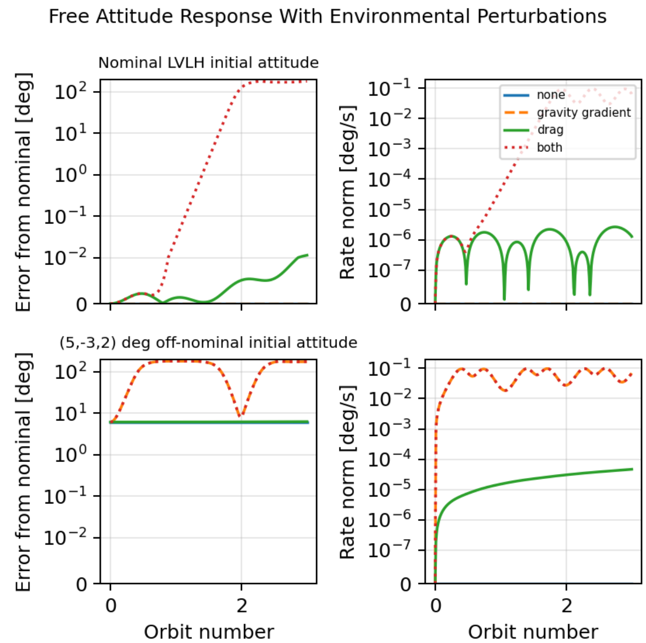

Fig. 5. Uncontrolled three-orbit attitude response with and without each perturbation. The ordinate uses a symmetric-log scale so the small drag response and the large gravity-gradient response are both visible.

III. Attitude Regulation

For Q3 I designed a continuous-time linear-quadratic regulator (LQR) to hold the same earth-pointing LVLH attitude used in Sec. II. The controller state is

$$\boldsymbol{x} = \begin{bmatrix} \delta\boldsymbol{\theta}^\top & \boldsymbol{\omega}^\top \end{bmatrix}^\top,$$

where $\delta\boldsymbol{\theta}$ is the small attitude-error rotation vector from the nominal LVLH attitude and $\boldsymbol{\omega}$ is the body angular velocity relative to the LVLH frame, expressed in the body frame. The linearized torque-input model is

$$\dot{\boldsymbol{x}} = \begin{bmatrix} 0_{3\times3} & I_3 \\ 0_{3\times3} & 0_{3\times3} \end{bmatrix} \boldsymbol{x} + \begin{bmatrix} 0_{3\times3} \\ J^{-1} \end{bmatrix} \boldsymbol{\tau}. \tag{10}$$

I used Bryson-style weights corresponding to $30\,\mathrm{deg}$ attitude error, $0.05\,\mathrm{deg/s}$ rate error, and $250\,\mathrm{N\,m}$ control torque. Solving the continuous algebraic Riccati equation gives the body-frame control law

$$\boldsymbol{\tau}_c = -K\boldsymbol{x},$$

with the diagonal gains in Table VI. Commands are limited to $(300, 300, 700)$ N·m in body roll, pitch, and yaw and then mapped to the four-CMG pyramid using the least-norm gimbal-rate solution

$$\dot{\boldsymbol{\theta}}_g = A(\boldsymbol{\theta}_g)^\dagger \boldsymbol{\tau}_c,$$

with the gimbal-rate limit from Sec. I-A.

TABLE VI
LQR GAINS FOR $\boldsymbol{\tau}_c = -K[\delta\boldsymbol{\theta}^\top, \boldsymbol{\omega}^\top]^\top$.

Axis	K_θ [N m/rad]	K_ω [N m s/rad]
Roll	477.5	4.52×10^5
Pitch	477.5	4.29×10^5
Yaw	477.5	5.23×10^5

A. Perfect-State Initial-Condition Tests

The first test uses perfect state feedback and no environmental disturbances, so it isolates the nonlinear attitude-regulation behavior from estimator and disturbance effects. I generated eight random attitude errors with magnitudes from 15° to 90° and small initial body rates. All cases converged to the nominal earth-pointing attitude within two orbits. The worst settling time to remain below $0.1\,\mathrm{deg}$ attitude error and $0.01\,\mathrm{deg/s}$ rate was $8800\,\mathrm{s}$ ($1.58\,\mathrm{orbits}$), and the largest final attitude error was $0.017\,\mathrm{deg}$. The largest achieved CMG torque was $479\,\mathrm{N\,m}$.

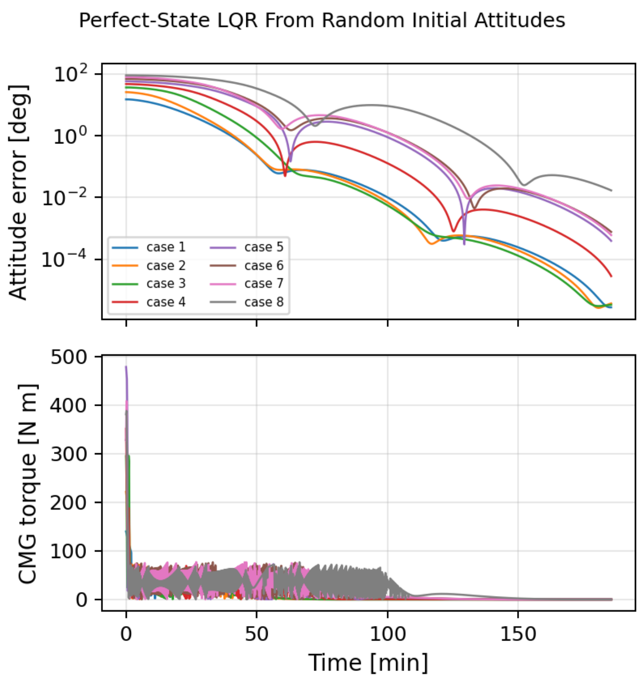

Fig. 6. Perfect-state LQR response from eight random initial attitude errors up to 90°. The CMGs saturate briefly for the largest slews, but all cases settle within the two-orbit simulation window.

B. Disturbed Orbit With Perfect State

Next I enabled gravity gradient and drag and initialized the spacecraft at $(2^\circ, -1^\circ, 1^\circ)$ from the nominal earth-pointing attitude. With perfect state feedback, the three-orbit simulation had a final attitude error of $0.027\,\mathrm{deg}$. The RMS nadir-pointing error over the last orbit was $0.070\,\mathrm{deg}$, and the maximum controller torque was only $20.4\,\mathrm{N\,m}$. This shows that once the large initial error is removed, the LQR torque needed to reject the Q2 environmental disturbances is small compared with CMG authority.

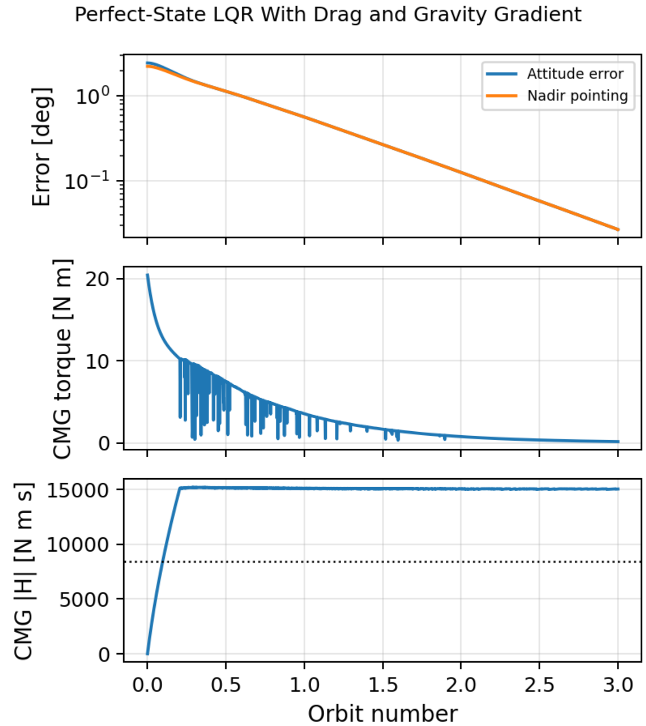

Fig. 7. Perfect-state LQR hold with gravity-gradient torque and drag enabled. The bottom panel shows the simple CMG steering law's stored momentum; no momentum-management null motion or desaturation is added in this controller baseline.

C. HW3 Estimator in the Loop

Finally, I replaced the perfect attitude and rate state with the HW3 MEKF estimate. The truth simulation uses the HW3 gyro random-walk model, star-tracker quaternion measurements, and magnetometer, sun, and horizon vector measurements. The controller receives the MEKF ECI-frame attitude estimate, converts it to the current LVLH frame, and subtracts the LVLH frame rate from the bias-corrected gyro measurement before applying the same LQR gain.

The estimator-in-the-loop case remains stable for the same three-orbit disturbed simulation. The last-orbit RMS nadir-pointing error is $0.092\,\mathrm{deg}$; the last-orbit RMS MEKF attitude error is $0.0061\,\mathrm{deg}$, with a maximum estimator attitude error of $0.018\,\mathrm{deg}$. The sensor noise therefore increases pointing error only slightly relative to the perfect-state baseline for this disturbance level.

a) Limitations.: The CMG steering law used here is a least-norm torque map, not a full ISS singularity-avoidance or momentum-management steering law. Large initial slews drive the modeled CMG momentum above the conservative inscribed sphere, although still inside the sampled full momentum envelope from Sec. I-A. A flight-quality implementation would add null-motion steering and periodic desaturation with the RCS; this is separate from the LQR state feedback design tested here.

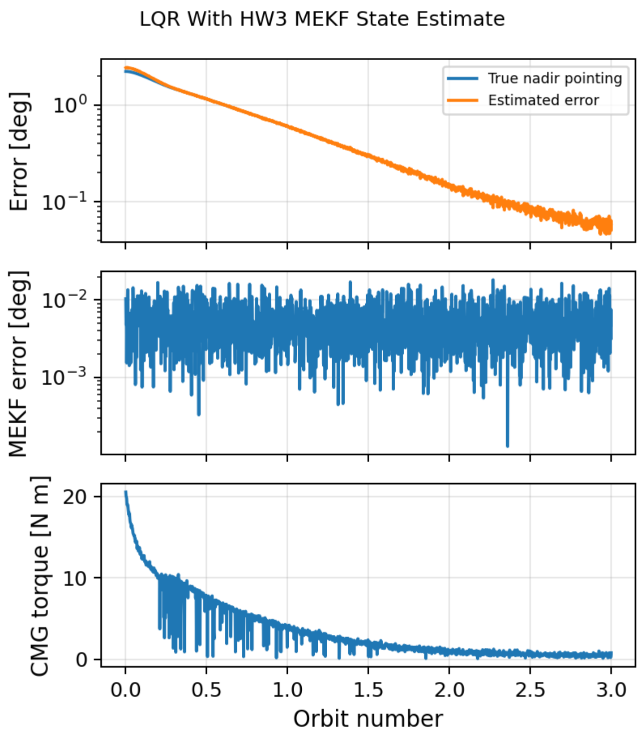

Fig. 8. LQR attitude hold using the HW3 MEKF state estimate. The estimated control error tracks the true nadir-pointing error closely; the middle panel shows the MEKF attitude error relative to truth.

IV. Eigen-Axis Slew

For Q4 I implemented an eigen-axis slew planner in `ISS/hw4/eigen_axis_slew.py`. The planner follows the Lecture 16 construction [5]: compute the relative attitude

$$R_e = R_0^\top R_f = \exp(\phi_f \hat{\boldsymbol{e}}^\times),$$

then move from the initial attitude R_0 to the final attitude R_f by

$$R(t) = R_0 \exp(\phi(t)\hat{\boldsymbol{e}}^\times). \tag{11}$$

I used the versine rest-to-rest profile

$$\phi(t) = \frac{\phi_f}{2}\left(1 - \cos\frac{\pi t}{T}\right), \qquad 0 \le t \le T, \tag{12}$$

with $\dot{\phi}(0) = \dot{\phi}(T) = 0$. The nominal body-frame angular velocity and acceleration are therefore

$$\boldsymbol{\omega}_d = \dot{\phi}\,\hat{\boldsymbol{e}}, \qquad \dot{\boldsymbol{\omega}}_d = \ddot{\phi}\,\hat{\boldsymbol{e}}.$$

The inverse-dynamics feedforward torque is the rigid-body Euler torque

$$\boldsymbol{\tau}_{\text{ff}} = J\dot{\boldsymbol{\omega}}_d + \boldsymbol{\omega}_d \times J\boldsymbol{\omega}_d. \tag{13}$$

A. Actuator Choice and Tracking Law

The requested test is a 180° maneuver. I chose an initial attitude that is 180° away from the nominal LVLH earth-pointing attitude by a pitch-axis flip, then slewed back to the nominal attitude. The resulting eigen-axis in the initial body frame is $-\hat{\boldsymbol{y}}_b$. With a one-orbit maneuver time $T = 5563\,\text{s} = 92.72\,\text{min}$, the nominal maximum rate is $0.0508\,\text{deg/s}$ and the inverse-dynamics torque peaks at only $53.6\,\text{N m}$.

Peak torque alone is not the limiting actuator constraint. The same one-orbit maneuver requires

$$\max_t \|J\boldsymbol{\omega}_d(t)\| = 9.49 \times 10^4\,\text{N m s}$$

of spacecraft angular momentum, which is much larger than both the sampled CMG full envelope from Sec. I-A ($4h = 1.904 \times 10^4\,\text{N m s}$) and the conservative inscribed sphere ($8454\,\text{N m s}$). A CMG-only one-orbit 180° slew would therefore violate the momentum capability even though the torque demand looks small. For this Q4 test I used the Zvezda RCS attitude-thruster couples from Sec. I-B, keeping each thruster below its $130\,\text{N}$ limit and strongly penalizing net force in the allocator.

The tracking controller uses the Q3 LQR gain, scaled by a factor of two, around the time-varying reference:

$$\boldsymbol{\tau}_c = \boldsymbol{\tau}_{\text{ff}} - 2K \begin{bmatrix} \delta\boldsymbol{\theta} \\ \delta\boldsymbol{\omega} \end{bmatrix}, \tag{14}$$

where $\delta\boldsymbol{\theta} = \log(R_d^\top R)$ and $\delta\boldsymbol{\omega} = \boldsymbol{\omega} - \boldsymbol{\omega}_d$ are computed from the HW3 MEKF attitude and rate estimate. I clipped the body torque command to $(250, 200, 250)$ N·m and allocated thruster forces by the bounded least-squares problem

$$\min_{0 \le \boldsymbol{u} \le \boldsymbol{u}_{\max}} \left\| W\left(B\boldsymbol{u} - \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{\tau}_c \end{bmatrix}\right)\right\|_2, \tag{15}$$

where B is the RCS wrench Jacobian from (6). The force rows of W are weighted 25 times more than the torque rows, so the solver prefers force-canceling thruster couples and accepts small torque error instead of using a large translational impulse.

B. Closed-Loop 180 Degree Slew

Figure 9 shows the MEKF-in-the-loop disturbed simulation with gravity gradient, drag, magnetic-field sensing, gyro noise, star tracker, sun sensor, horizon sensor, and magnetometer all enabled. The closed-loop trajectory follows the planned rest-to-rest profile and settles after $8670\,\text{s}$ ($1.56\,\text{orbits}$) to below $0.25\,\text{deg}$ attitude error and $0.01\,\text{deg/s}$ rate. The final attitude error after the three-orbit simulation is $0.020\,\text{deg}$. The maximum tracking error during the maneuver is $6.01\,\text{deg}$; this peak occurs near the middle of the slew, where gravity-gradient torque is largest and the controller is trading tracking tightness against the conservative RCS torque limit.

The actuator commands remain inside the modeled RCS capability (Fig. 10). The largest commanded body torque is $125.0\,\text{N m}$, the largest achieved torque is $125.0\,\text{N m}$, and the largest individual thruster command is $78.7\,\text{N}$. The allocation keeps net force below $3.3 \times 10^{-2}\,\text{N}$, so the maneuver is effectively torque-only in the coupled orbit simulation. The total absolute thruster impulse, computed as $\int \sum_i |u_i|\,dt$, is $6.17 \times 10^5\,\text{N s}$; this is not a propellant-optimal steering law, only a bounded inverse-dynamics and feedback test.

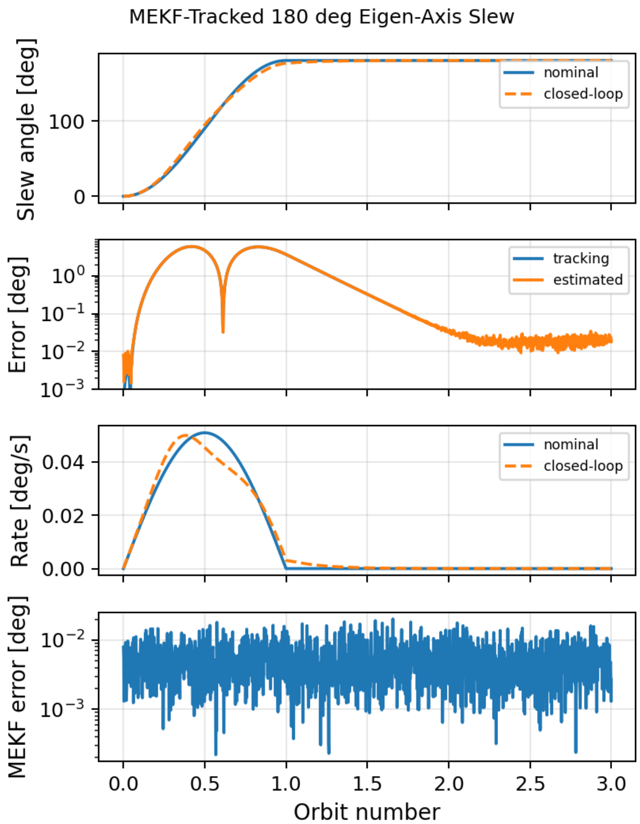

Fig. 9. Nominal and closed-loop states for the 180° eigen-axis slew. The controller uses the HW3 MEKF state estimate; the estimator error remains small compared with the trajectory-tracking error.

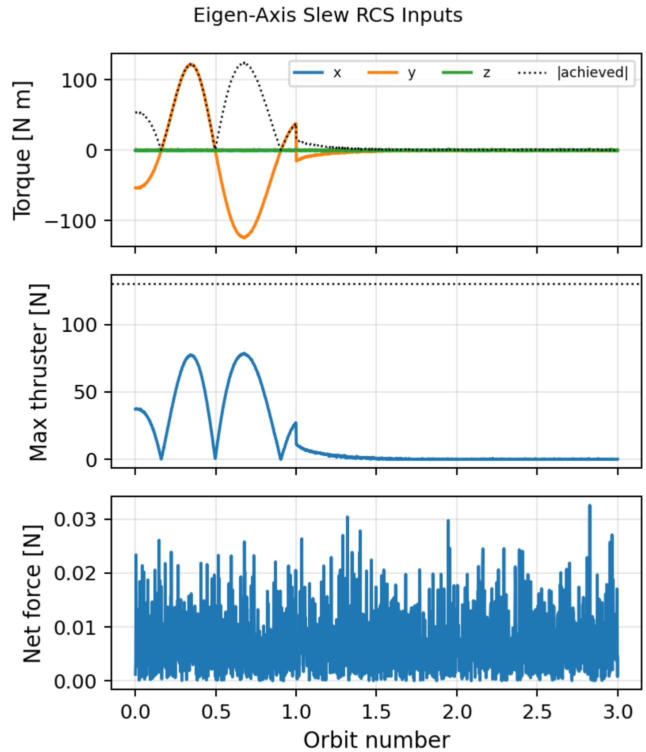

Fig. 10. RCS torque and force commands during the eigen-axis slew. The dotted line in the middle panel is the 130 N per-thruster limit.

C. Comparison With Direct Regulation

I compared the planned slew against the same LQR feedback law used as a direct regulator from the identical 180° initial attitude (Fig. 11). In this single-axis case the direct regulator converges slightly faster, settling in $7430\,\mathrm{s}$ and ending at $0.0167\,\mathrm{deg}$ error, because it simply saturates the pitch torque command near $200\,\mathrm{N\,m}$ and drives toward the target as fast as the imposed torque limit permits. It also uses a larger individual thruster command, $112.8\,\mathrm{N}$, and reaches a much higher rate error, $0.136\,\mathrm{deg/s}$, than the planned slew's $0.0085\,\mathrm{deg/s}$.

The practical difference is predictability. The eigen-axis slew gives a specified maneuver time, a known rest-to-rest rate profile, and bounded inverse-dynamics inputs before simulation. The direct regulator has no planned rate or torque history; for a far initial condition it behaves like a saturated nonlinear controller. It works for this pitch-axis test because the attitude-log error selects the right rotation axis and the RCS torque limit is high enough, but it does not by itself guarantee actuator-feasible motion for arbitrary large-angle slews.

References

[1] B. Wie, *Space Vehicle Dynamics and Control*, 2nd ed. AIAA Education Series, 2008. Chapter 7.

[2] N. Bedrossian, S. Bhatt, M. Lebsock, and E. Wong, "International Space Station zero-propellant maneuver," *Journal of Guidance, Control, and Dynamics*, vol. 32, no. 4, pp. 1193–1201, 2009.

[3] Honeywell Defense and Space, "M1000 Control Moment Gyroscope," product information sheet.

[4] NASA, "International Space Station Propulsion System," ISS Familiarization Manual, ISS FAM C TD9702A, 1998.

[5] Z. Manchester, "Lecture 16: Attitude regulation and slew maneuvers," Spacecraft Attitude Dynamics and Control lecture notes, 2026. Available: https://github.com/zacmanchester/spacecraft-attitude-course/blob/main/lecture%20notes/Lecture%2016/Lecture%2016.pdf

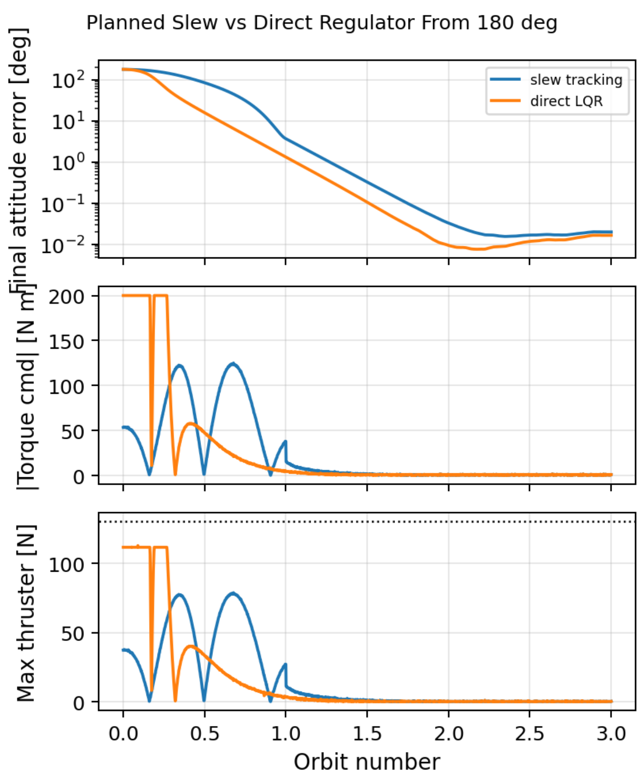

Fig. 11. Comparison between the planned eigen-axis slew and a direct LQR regulator from the same $180°$ initial attitude. The direct regulator is faster here, but only by saturating the torque command and using higher rates.